

2.5 The Conjugate Normal Equations

The least squares estimate of an estimable function $\underline{\lambda}'\underline{\beta}$ can be obtained by computing a solution $\widehat{\underline{\beta}}$ to the normal equations and by then forming $\underline{\lambda}'\widehat{\underline{\beta}}$, which is unique and can be written as $\underline{\lambda}'(X'X)^-X'y$.

Alternatively, we can compute a solution $\underline{r}$ to the conjugate normal equations

$$X'X\underline{r} = \underline{\lambda}$$

and obtain the least squares estimate of $\underline{\lambda}'\underline{\beta}$ as $\underline{r}'X'y$.

Theorem 2.7. *Under Models II—VII, $\underline{\lambda}'\underline{\beta}$ is estimable if and only if there exists an $\underline{r}$ such that $X'X\underline{r} = \underline{\lambda}$; that is, if and only if the conjugate normal equations are consistent.*

Corollary. Under Models II—VII, $\underline{\lambda}'\underline{\beta}$ is estimable if and only if $X'X(X'X)^-\underline{\lambda} = \underline{\lambda}$.

Theorem 2.8. *Under Models II—VII, the least squares estimate of an estimable function $\underline{\lambda}'\underline{\beta}$ is equal to $\underline{r}'X'y$, where $\underline{r}$ is any solution to $X'X\underline{r} = \underline{\lambda}$.*

2.6 Expectations/Variances/Covariances of Least Squares Estimators

Suppose that $\underline{\lambda}' \in \mathcal{R}(X)$ so that $\underline{\lambda}'\underline{\beta}$ is estimable under Models II—VII. Taking $\underline{r}$ to be any solution to the conjugate normal equations

$$X'X\underline{r} = \underline{\lambda},$$

the ordinary least squares estimator of $\underline{\lambda}'\underline{\beta}$ is then $\underline{a}'\underline{y}$, where $\underline{a}' = \underline{r}'X'$ and also

$$\underline{a}'X = \underline{r}'X'X = \underline{\lambda}'.$$

Under Models II—VII, the ordinary least squares estimator of $\underline{\lambda}'\underline{\beta}$ is an unbiased estimator of the estimable function $\underline{\lambda}'\underline{\beta}$.

For two random vectors $\underline{x} = (x_1, \dots, x_m)'$ and $\underline{z} = (z_1, \dots, z_n)'$, we write $\text{cov}(\underline{x}, \underline{z})$ for the $m \times n$ matrix whose (i, j) th element is the covariance between x_i and z_j . That is,

$$\text{cov}(\underline{x}, \underline{z}) = \begin{bmatrix} \text{cov}(x_1, z_1) & \cdots & \text{cov}(x_1, z_n) \\ \vdots & \vdots & \vdots \\ \text{cov}(x_m, z_1) & \cdots & \text{cov}(x_m, z_n) \end{bmatrix}.$$

The matrix $\text{Var}(\underline{x}) = \text{cov}(\underline{x}, \underline{x})$ is called the variance-covariance matrix, or simply the covariance matrix, of $\underline{x}$.

Lemma 2.4. *If $\underline{x}$ is a random vector and L and M are matrices of constants having the same number of rows as $\underline{x}$ has elements, then*

$$\text{cov}(L'\underline{x}, M'\underline{x}) = L'\Sigma M, \text{ where } \Sigma = \text{Var}(\underline{x}).$$

Corollary. $\text{Var}(L'\underline{x}) = L'\Sigma L$, where $\Sigma = \text{Var}(\underline{x})$.

Definition. For a matrix of constants Λ having p rows, the vector $\Lambda' \underline{\beta}$ is said to be estimable if each of its elements is estimable.

Suppose that $\Lambda_1' \underline{\beta}$ and $\Lambda_2' \underline{\beta}$ (where Λ_1 and Λ_2 are matrices of constants having p rows each) are both estimable. Let $\widehat{\underline{\beta}}$ represent any solution to the normal equations, and take R_1 and R_2 to be any two matrices satisfying

$$X'XR_1 = \Lambda_1 \quad \text{and} \quad X'XR_2 = \Lambda_2.$$

The least squares estimators of $\Lambda_1' \underline{\beta}$ and $\Lambda_2' \underline{\beta}$, that is, the vectors whose elements are the ordinary least squares estimators of the corresponding elements of $\Lambda_1' \underline{\beta}$ and $\Lambda_2' \underline{\beta}$, are given by

$$\Lambda_1' \widehat{\underline{\beta}} = R_1' X' \underline{y} \quad \text{and} \quad \Lambda_2' \widehat{\underline{\beta}} = R_2' X' \underline{y}.$$

Under Model VII, by Lemma 1.4 and its Corollary 1 (ii),

$$\begin{aligned} \text{cov}(\Lambda_1' \widehat{\underline{\beta}}, \Lambda_2' \widehat{\underline{\beta}}) &= \text{cov}(R_1' X' \underline{y}, R_2' X' \underline{y}) \\ &= R_1' X' V X R_2 \\ &= R_1' [X' X (X' X)^{-1} X'] V [X (X' X)^{-1} X'] R_2 \\ &= (R_1' X' X) (X' X)^{-1} X' V X (X' X)^{-1} (X' X R_2) \\ &= \Lambda_1' (X' X)^{-1} X' V X (X' X)^{-1} \Lambda_2, \end{aligned}$$

and, in particular,

$$\text{Var}(\Lambda_1' \widehat{\underline{\beta}}) = \Lambda_1' (X' X)^{-1} X' V X (X' X)^{-1} \Lambda_1.$$

Under Model II (the Gauss-Markov model), which can be regarded as a special case of Model VII,

$$\begin{aligned} \text{cov}(\Lambda_1' \widehat{\underline{\beta}}, \Lambda_2' \widehat{\underline{\beta}}) &= R_1' X' (\sigma^2 I) X R_2 = \sigma^2 R_1' X' X R_2 = \sigma^2 R_1' [X' X (X' X)^{-1} X'] R_2 \\ &= \sigma^2 (R_1' X' X) (X' X)^{-1} (X' X R_2) = \sigma^2 \Lambda_1' (X' X)^{-1} \Lambda_2, \end{aligned}$$

and, in particular,

$$\text{Var}(\Lambda_1' \widehat{\underline{\beta}}) = \sigma^2 \Lambda_1' (X' X)^{-1} \Lambda_1.$$

2.7 The Gauss-Markov Theorem

Under Models II—VII, the (ordinary) least squares estimator of an estimable function $\underline{\lambda}'\underline{\beta}$ is unbiased. Moreover, under Model II, the least squares estimator of an estimable function $\underline{\lambda}'\underline{\beta}$ is the best (minimum variance) linear unbiased estimator (BLUE) as described by the following Gauss-Markov theorem.

Theorem 2.9. (The Gauss-Markov Theorem) *Under Models II—VII, if $\underline{\lambda}'\underline{\beta}$ is estimable and $\widehat{\underline{\beta}}$ is any solution to the normal equations, then $\underline{\lambda}'\widehat{\underline{\beta}}$ is a linear unbiased estimator of $\underline{\lambda}'\underline{\beta}$. Under Model II, the variance of $\underline{\lambda}'\widehat{\underline{\beta}}$ is uniformly less than that of any other linear unbiased estimator of $\underline{\lambda}'\underline{\beta}$, that is, $\underline{\lambda}'\widehat{\underline{\beta}}$ is the best (minimum variance) linear unbiased estimator (BLUE) of $\underline{\lambda}'\underline{\beta}$.*

Proof. If $c + \underline{a}'\underline{y}$ is any linear unbiased estimator of $\underline{\lambda}'\underline{\beta}$, then $c = 0$, $\underline{\lambda}' = \underline{a}'X$ (Lemma 2.1); and

$$\begin{aligned}\text{Var}(c + \underline{a}'\underline{y}) &= \text{Var}(\underline{a}'\underline{y}) = \text{Var}[\underline{\lambda}'\widehat{\underline{\beta}} + (\underline{a}'\underline{y} - \underline{\lambda}'\widehat{\underline{\beta}})] \\ &= \text{Var}(\underline{\lambda}'\widehat{\underline{\beta}}) + \text{Var}(\underline{a}'\underline{y} - \underline{\lambda}'\widehat{\underline{\beta}}) + 2\text{cov}(\underline{\lambda}'\widehat{\underline{\beta}}, \underline{a}'\underline{y} - \underline{\lambda}'\widehat{\underline{\beta}}).\end{aligned}$$

Finally, if $\text{Var}(c + \underline{a}'\underline{y}) = \text{Var}(\underline{\lambda}'\widehat{\underline{\beta}})$ ever holds, it must be that $\text{Var}(\underline{a}'\underline{y} - \underline{\lambda}'\widehat{\underline{\beta}}) = 0$ or

$$\begin{aligned}\text{Var}(\underline{a}'\underline{y} - \underline{\lambda}'\widehat{\underline{\beta}}) &= \text{Var}(\underline{a}'\underline{y} - \underline{r}'X'\underline{y}) = \text{Var}[(\underline{a} - X\underline{r})'\underline{y}] \\ &= (\underline{a} - X\underline{r})'(\sigma^2 I)(\underline{a} - X\underline{r}) = \sigma^2(\underline{a} - X\underline{r})'(\underline{a} - X\underline{r}) = 0 \iff \underline{a} - X\underline{r} = \underline{0} \\ &\iff \underline{a}' = \underline{r}'X' \iff \underline{a}'\underline{y} = \underline{\lambda}'\widehat{\underline{\beta}} \text{ for all } \underline{y} \iff c + \underline{a}'\underline{y} = \underline{\lambda}'\widehat{\underline{\beta}} \text{ for all } \underline{y}.\end{aligned}$$

That is, $\underline{\lambda}'\widehat{\underline{\beta}}$ is the unique best linear unbiased estimator of $\underline{\lambda}'\underline{\beta}$.

Suppose now that we wish to simultaneously estimate k estimable functions of $\underline{\beta}$.

- Choose the $p \times k$ matrix Λ in such a way that $\Lambda'\underline{\beta}$ is the vector whose elements are the k estimable functions.
- Let $\widehat{\underline{\beta}}$ be any solution to the normal equations.
- Then, the estimator $\Lambda'\widehat{\underline{\beta}}$ is one unbiased estimator of $\Lambda'\underline{\beta}$.

We might ask: In what sense (or senses) is $\Lambda'\widehat{\underline{\beta}}$ a “good” estimator?

To answer this, we consider some matrix properties regarding the variance of $\Lambda'\widehat{\underline{\beta}}$.

We'll first introduce some definitions.

Definitions. If $\underline{x}$ is a column vector of r real variables and A is an $r \times r$ matrix of constants, then we refer to the function

$$q(\underline{x}) = \underline{x}' A \underline{x}$$

as a quadratic form, where quadratic form $q(\underline{x})$ is said to be

nonnegative definite if $q(\underline{x}) > 0$ for all real vectors $\underline{x}$;

positive definite if $q(\underline{x}) > 0$ for all nonzero real vectors $\underline{x}$;

positive semidefinite if it is nonnegative definite but not positive definite (i.e., $q(\underline{x}) \geq 0$ with equality holding for at least one nonzero real vector $\underline{x}$)

A symmetric square matrix A is nonnegative definite, positive definite, positive semidefinite according to whether quadratic form $\underline{x}' A \underline{x}$ is nonnegative definite, positive definite, or positive semidefinite, respectively.

Theorem 2.10. *Under Model II, suppose that $\Lambda' \underline{\beta}$ is any k -dimensional estimable vector and that $\underline{c} + A' \underline{y}$ is any vector of linear unbiased estimators of the elements of $\Lambda' \underline{\beta}$. Then, the matrix $\text{Var}(\underline{c} + A' \underline{y}) - \text{Var}(\Lambda' \widehat{\underline{\beta}})$ is nonnegative definite, where $\widehat{\underline{\beta}}$ is any solution to the normal equations.*

Proof. For any $k \times 1$ vector $\underline{\ell}$, by the corollary to Lemma 2.4,

$$\underline{\ell}' [\text{Var}(\underline{c} + A' \underline{y}) - \text{Var}(\Lambda' \widehat{\underline{\beta}})] \underline{\ell} = \underline{\ell}' \text{Var}(\underline{c} + A' \underline{y}) \underline{\ell} - \underline{\ell}' \text{Var}(\Lambda' \widehat{\underline{\beta}}) \underline{\ell} = \text{Var}[\underline{\ell}' (\underline{c} + A' \underline{y})] - \text{Var}(\underline{\ell}' \Lambda' \widehat{\underline{\beta}}).$$

By Cor 2/Theorem 2.1, $\tilde{\underline{\lambda}}' \underline{\beta}$ for $\tilde{\underline{\lambda}} \equiv \underline{\ell}' \Lambda'$ is estimable with least squares estimator as $\tilde{\underline{\lambda}}' \widehat{\underline{\beta}}$. Since $\underline{\ell}' (\underline{c} + A' \underline{y})$ is a linear unbiased estimator of $\tilde{\underline{\lambda}}' \underline{\beta} = (\underline{\ell}' \Lambda') \underline{\beta}$,

Corollary 1. Under the condition of Theorem 2.10,

$$\text{tr}[\text{Var}(\underline{c} + A' \underline{y})] \geq \text{tr}[\text{Var}(\Lambda' \widehat{\underline{\beta}})].$$

Corollary 2. Under the condition of Theorem 2.10,

$$\det[\text{Var}(\underline{c} + A' \underline{y})] \geq \det[\text{Var}(\Lambda' \widehat{\underline{\beta}})].$$

Definition. The determinant of the variance-covariance matrix of an estimator is referred to as the generalized variance of the estimator.

Remark. (This material is optional.) To prove Corollary 2, we use Theorem 12.2.14 (7) in the first edition of Graybill's book *Introduction to Matrices with Applications in Statistics*, as stated in Result 1 below, which extends to Result 2 below. Corollary 2 follows from Result 2 with

$$U = \text{Var}(\underline{c} + A' \underline{y}), \quad V = \text{Var}(\underline{c} + A' \underline{y}) - \text{Var}(\Lambda' \widehat{\underline{\beta}}), \quad \text{and} \quad U - V = \text{Var}(\Lambda' \widehat{\underline{\beta}}).$$

Result 1. For $m \times m$ matrices U and V , if U is positive definite and V and $U - V$ are nonnegative definite, then $\det(U) \geq \det(U - V)$.

Result 2. For $m \times m$ matrices U and V , if U , V , and $U - V$ are nonnegative definite, then $\det(U) \geq \det(U - V)$.

Proof. If U is positive definite, Result 2 becomes Result 1. If U is positive semidefinite, then there exists a nonzero vector $\underline{t}$ such that $\underline{t}' U \underline{t} = 0$. Thus, $0 = \underline{t}' U \underline{t} = \underline{t}' [V + (U - V)] \underline{t} = \underline{t}' V \underline{t} + \underline{t}' (U - V) \underline{t}$, implying $\underline{t}' (U - V) \underline{t} = 0$ as V and $U - V$ are nonnegative definite. Then, U and $U - V$ are positive semidefinite (not full rank) and $\det(U) = 0 = \det(U - V)$ hold. $\square$

2.8 Estimation of the Residual Variance

We now consider the estimation of σ^2 for Model II. We begin by introducing some lemmas that will be useful in this and subsequent discussions.

Definition. A square matrix A is said to be an idempotent matrix if $A^2 = A$.

Lemma 2.5. For any square matrix A such that $A^2 = mA$ for some scalar m ,

$$\text{tr}(A) = m\text{rank}(A).$$

Proof. This holds when $A = 0$. For $A \neq 0$, $r = \text{rank}(A) \geq 1$. By the proof of Theorem 1.1, there exist nonsingular matrices P and Q such that

$$PAQ = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix},$$

where I_r is the $r \times r$ identity matrix. We have

$$\begin{aligned} A^2 = mA &\iff \left(P^{-1} \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} Q^{-1} \right) \left(P^{-1} \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} Q^{-1} \right) = m \left(P^{-1} \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} Q^{-1} \right) \\ &\iff \left(\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} Q^{-1} \right) \left(P^{-1} \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} \right) = m \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}. \end{aligned}$$

Note that for any $p \times q$ matrix F and $q \times p$ matrix G , $\text{tr}(FG) = \text{tr}(GF)$.

Thus,

$$\begin{aligned} m\text{rank}(A) &= \text{tr} \left(m \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} \right) = \text{tr} \left\{ \left(\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} Q^{-1} \right) \left(P^{-1} \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} \right) \right\} \\ &= \text{tr} \left\{ \left(P^{-1} \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} \right) \left(\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} Q^{-1} \right) \right\} = \text{tr} \left(P^{-1} \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} Q^{-1} \right) = \text{tr}(A). \end{aligned}$$

□

Corollary. If A is idempotent, then $\text{tr}(A) = \text{rank}(A)$.

Recall the following properties of the projection matrix $P_X = XB$ where B is a solution to $X'XB = X'$ (after Lemma 1.5):

- (1) P_X is invariant to the choice of solution B ,
- (2) P_X is symmetric,
- (3) $P_X = X(X'X)^-X'$,
- (4) $P_X X = X$,
- (5) P_X is idempotent, that is, $P_X^2 = P_X$.

It follows that $(I - P_X)$ is also symmetric and idempotent.

Lemma 2.6. *For any matrix A , P_A and $I - P_A$ are both symmetric idempotent matrices, where $P_A = A(A'A)^-A'$.*

Lemma 2.7. *For any $n \times p$ matrix A , $\text{rank}(P_A) = \text{rank}(A)$ and $\text{rank}(I - P_A) = n - \text{rank}(A)$.*

Lemma 2.8. *If $\underline{x}$ is an $n \times 1$ random vector having mean vector $\underline{\mu}$ and variance-covariance matrix Σ and if A is an $n \times n$ matrix of constants, then*

$$E(\underline{x}' A \underline{x}) = \text{tr}(A \Sigma) + \underline{\mu}' A \underline{\mu}.$$

Proof.

$$\begin{aligned} E(\underline{x}' A \underline{x}) &= E[\text{tr}(\underline{x}' A \underline{x})] = E[\text{tr}(A \underline{x} \underline{x}')] = \text{tr}[E(A \underline{x} \underline{x}')] = \text{tr}[A E(\underline{x} \underline{x}')] \\ &= \text{tr}[A(\Sigma + \underline{\mu} \underline{\mu}')] = \text{tr}(A \Sigma) + \text{tr}(A \underline{\mu} \underline{\mu}') = \text{tr}(A \Sigma) + \text{tr}(\underline{\mu}' A \underline{\mu}) = \text{tr}(A \Sigma) + \underline{\mu}' A \underline{\mu}. \end{aligned}$$

□

Let $\underline{\hat{\beta}}$ be any solution to the normal equations and take the sum of squares of the deviations of the elements of $\underline{y}$ from the elements of $X\underline{\hat{\beta}}$ as

$$Q(\underline{\hat{\beta}}) = (\underline{y} - X\underline{\hat{\beta}})'(\underline{y} - X\underline{\hat{\beta}}) = (\underline{y} - P_X \underline{y})'(\underline{y} - P_X \underline{y}) = [(I - P_X)\underline{y}]'[(I - P_X)\underline{y}] = \underline{y}'(I - P_X)\underline{y}.$$

Under Model II, we have $E[Q(\underline{\hat{\beta}})] =$

Thus, under Model II,

$$\hat{\sigma}^2 = (\underline{y} - X\underline{\hat{\beta}})'(\underline{y} - X\underline{\hat{\beta}})/(n - p^*) = \underline{y}'(I - P_X)\underline{y}/(n - p^*) = \underline{y}'(\underline{y} - X\underline{\hat{\beta}})/(n - p^*)$$

is an unbiased estimator of σ^2 .